

1. Determine the multiplicative inverse modulo 26 for each of the following integers.

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 mod 26

a) $1^{-1} \bmod 26$

$$26 = 1(25) + 1$$

$$26 + 1(-25) = 1$$

$$26 - 25 = 1$$

$$1^{-1} \bmod 26 = 1 \bmod 26 = \underline{\underline{1}}$$

b) $3^{-1} \bmod 26$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$3(9) + 26(-1) = 1$$

$$3^{-1} = 9$$

$$3^{-1} \bmod 26 = 9 \bmod 26 = \underline{\underline{9}}$$

c) $5^{-1} \bmod 26$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$5^{-1} \bmod 26 = 21 \bmod 26 = \underline{\underline{21}}$$

$$d) 7^{-1} \bmod 26$$

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[26 + 7(-3)] + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-3) + 7(-2) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} \bmod 26 = 15 \bmod 26 = \underline{\underline{15}}$$

$$e) 9^{-1} \bmod 26$$

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

$$9^{-1} \bmod 26$$

$$= 3 \bmod 26$$

$$= \underline{\underline{3}}$$

$$f) 11^{-1} \bmod 26$$

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$11^{-1} \bmod 26 = 19 \bmod 26 = \underline{\underline{19}}$$

$$g) 15^{-1} \bmod 26$$

$$26 = 15(1) + 11$$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + [11 + 4(-2)](-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + [26 + 15(-1)](-2) + [15 + 11(-1)](2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + [26 + 15(-1)](-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(1) + 26(-4) = 1$$

$$15^{-1} = 1 + (5)11 = 11$$

$$15^{-1} \bmod 26 = 1 \bmod 26 = \underline{\underline{1}}$$

h) $17^{-1} \bmod 26$

$$26 = 17(1) + 9$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + [17 + 9(-1)](-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + [26 + 17(-1)](1) = 1$$

$$26 + 17(-2) + 26(1) + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$17^{-1} \bmod 26 = 23 \bmod 26 = \underline{\underline{23}}$$

i) $19^{-1} \bmod 26$

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$19 + 7(-2) + [7 + 5(-1)](-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + [26 + 19(-1)](-4) + [19 + 7(-2)](2) = 1$$

$$19 + 26(-4) + 19(4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + [26 + 19(-1)](-4) = 1$$

$$19(7) + 26(-4) + 26(-4) + 19(4) = 1$$

$$26(-8) + 19(11) = 1$$

$$19^{-11} = 11$$

$$19^{-1} \bmod 26 = 11 \bmod 26 = \underline{\underline{11}}$$

$$j) 21^{-1} \bmod 26$$

$$26 = 21(1) + 5$$

$$21 = 5(4) + 1$$

$$21 + 5(-4) = 1$$

$$21 + [26 + 21(-1)](-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$26(-4) + 21(5) = 1$$

$$21^{-1} = 5$$

$$21^{-1} \bmod 26 = 5 \bmod 26 = \underline{\underline{5}}$$

$$k) 23^{-1} \bmod 26$$

$$26 = 23(1) + 3$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + [23 + 3(-7)](-1) = 1$$

$$26 + 23(-1) + 23(-1) + 3(7) = 1$$

$$26 + 23(-2) + [26 + 23(-1)](7) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$23^{-1} \bmod 26 = 17 \bmod 26 = \underline{\underline{17}}$$

$$1) 25^{-1} \bmod 26$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$25^{-1} \bmod 26 = 25 \bmod 26 = \underline{\underline{25}}$$

2) Choose any string of your choice to encrypt using the affine cipher with key values : multiplier 5 and additive 8.

Plaintext : PRANEETHA

Multiplier = $K_1 = 5$

Additive = $K_2 = 8$

Formula : $C = (P \cdot K_1 + K_2) \bmod 26$

$$P = (15(5) + 8) \bmod 26 = 83 \bmod 26 = 5 = F$$

$$R = (17(5) + 8) \bmod 26 = 93 \bmod 26 = 15 = P$$

$$A = (0(5) + 8) \bmod 26 = 8 \bmod 26 = 8 = I$$

$$N = (13(5) + 8) \bmod 26 = 73 \bmod 26 = 21 = V$$

$$E = (4(5) + 8) \bmod 26 = 28 \bmod 26 = 2 = C$$

$$\begin{aligned}
 E &= (4(5) + 8) \bmod 26 = 28 \bmod 26 = 2 = C \\
 T &= (19(5) + 8) \bmod 26 = 103 \bmod 26 = 25 = Z \\
 H &= (7(5) + 8) \bmod 26 = 43 \bmod 26 = 17 = R \\
 A &= (0(5) + 8) \bmod 26 = 8 \bmod 26 = 8 = I
 \end{aligned}$$

$\therefore$ Ciphertext : **FPIVCCZRI**

3) Encrypt "HI" using the Affine cipher with $a=11$ and $b=5$.

$$\text{Formula: } C = (P \cdot a + b) \bmod 26$$

$$H = (7(11) + 5) \bmod 26 = 82 \bmod 26 = 4 = E$$

$$I = (8(11) + 5) \bmod 26 = 93 \bmod 26 = 15 = P$$

$\therefore$ Ciphertext : "EP"

4) The ciphertext VCR was encrypted using the affine function $9x + 2 \bmod 26$. Find the plaintext.

$$C = (9x + 2) \bmod 26$$

$$P = x = 9^{-1} (C - 2) \bmod 26$$

$$9^{-1} : 26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 26 + 9(-2)$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$26(-1) + 9(3) = 1$$

$$9^{-1} = 3$$

$$9^{-1} \bmod 26 = 3 \bmod 26 = 3$$

$$U = 3(20-2) \bmod 26 = 54 \bmod 26 = 2 = C$$

$$C = 3(2-2) \bmod 26 = 0 \bmod 26 = 0 = A$$

$$R = 3(17-2) \bmod 26 = 45 \bmod 26 = 19 = T$$

$\therefore$ Plaintext : CAT

5) Decrypt the ciphertext using Affine cipher, KDXJUVDEXZY using the key 3 (Multiplier) and 4.

$$\text{Multiplier} = a = 3$$

$$\text{Additive} = b = 4$$

$$\text{Formula: } P = a^{-1}(C - b) \bmod 26$$

$$P = 3^{-1}(C - 4) \bmod 26$$

$$3^{-1}: 26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + [26 + 3(-8)] - 1 = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$3(9) + 26(-1) = 1$$

$$3^{-1} = 9$$

$$3^{-1} \bmod 26 = 9 \bmod 26 = 9$$

$$K = 9 \quad (10 - 4) \bmod 26 = 54 \bmod 26 = 2 = C$$

$$D = 9 \quad (3 - 4) \bmod 26 = -9 \bmod 26 = 17 = R$$

$$Y = 9 \quad (24 - 4) \bmod 26 = 180 \bmod 26 = 24 = Y$$

$$X = 9 \quad (23 - 4) \bmod 26 = 171 \bmod 26 = 15 = P$$

$$J = 9 \quad (9 - 4) \bmod 26 = 45 \bmod 26 = 19 = T$$

$$V = 9 \quad (20 - 4) \bmod 26 = 144 \bmod 26 = 14 = O$$

$$W = 9 \quad (22 - 4) \bmod 26 = 162 \bmod 26 = 6 = G$$

$$D = 9 \quad (3 - 4) \bmod 26 = -9 \bmod 26 = 17 = R$$

$$E = 9 \quad (4 - 4) \bmod 26 = 0 \bmod 26 = 0 = A$$

$$X = 9 \quad (23 - 4) \bmod 26 = 171 \bmod 26 = 15 = P$$

$$Z = 9 \quad (25 - 4) \bmod 26 = 189 \bmod 26 = 7 = H$$

$$Y = 9 \quad (24 - 4) \bmod 26 = 180 \bmod 26 = 24 = Y$$

$\therefore$ Plaintext : CRYPTOGRAPHY

6) Use $K1 = 13$ multiplier, $K2 = 4$ (Even though it is not a valid key - try to encrypt)
Encrypt the string "input" and "alter" using the above key and find what is the issue.

Formula: $C = (P \cdot \text{Multi} + \text{Add}) \bmod 26$

$$\text{Multi} = K1 = 13$$

$$\text{Add} = K2 = 4$$

$$C = (P \cdot 13 + 4) \bmod 26$$

13 is not a valid key since 13 and 26 are not co-primes. We cannot find its inverse.

String = "input"

$$I = (8(13) + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13(13) + 4) \bmod 26 = 173 \bmod 26 = 17 = R$$

$$P = (15(13) + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$U = (20(13) + 4) \bmod 26 = 264 \bmod 26 = 4 = E$$

$$T = (19(13) + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

Ciphertext : ERRER

String = "alter"

$$A = (0(13) + 4) \bmod 26 = 4 \bmod 26 = 4 = E$$

$$L = (11(13) + 4) \bmod 26 = 147 \bmod 26 = 17 = R$$

$$T = (19(13) + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

$$E = (4(13) + 4) \bmod 26 = 56 \bmod 26 = 4 = E$$

$$R = (17(13) + 4) \bmod 26 = 225 \bmod 26 = 17 = R$$

Ciphertext : ERRER

The problem is that $K_1 = 13$ is not a valid multiplier key because $\gcd(13, 26) = 13 \neq 1$. Therefore, 13 has no multiplicative inverse modulo 26.

This causes different plaintext to produce the same ciphertext.

$\therefore$ K_1 must be coprime with 26 in an Affine cipher.

7) Find the inverse matrix for the key

$$\text{Key} \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$$

$$\det(K) = |K| = (7 \times 3) - (5 \times 2) = 11 \neq 0$$

$$11^{-1} : 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$(5 \times 3) - (4 \times 1) = 1 \quad 4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26(3) + 11(-1) = 1$$

$$26 - 1 = 19$$

$$11^{-1} \text{ mod } 26 = 19 \text{ mod } 26 = 19$$

Finding the adjoint of the matrix

$$\text{adj}(K) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \text{ is the inverse matrix}$$

8) Find the inverse of the following matrix whose entries are considered modulo 26:

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

$$K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

$$\text{Determinant of } K = |K| = (11 \times 3) - (13 \times 2) = 7$$

$$7^{-1} : 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$26 + 7(-3) + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-5) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} \text{ mod } 26 = 15 \text{ mod } 26 = 15$$

$$\text{Adjoint of } K = \text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix} \text{ is the inverse matrix.}$$

9) Is the key valid?

$$K = \begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

For the key to be valid, its determinant should not be equal to zero and the determinant must be coprime with 26.

$$\text{Determinant } K = |K| = (21 \times 2) - (13 \times 14) = -140$$

Condition 1 is satisfied i.e., $\det |K| = -140 \neq 0$

For condition 2,

$$-140 \pmod{26} = 16$$

$$\gcd(16, 26) = 2 \neq 1$$

16 is not coprime to 26 and it doesn't have inverse.

$\therefore$ Condition 2 fails

$\therefore$ The key is not valid.

10) Encrypt using 2×2 Hill Cipher for the plaintext "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

Plaintext : S O | L V | E D

$$\text{Key, } K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$$

$$|K| = 3(11) - 7(8) = -23 \bmod 26 = 3$$

3 is coprime to 26.

$$\text{Formula : } P = (C \times K) \bmod 26$$

~~$$SO = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \bmod 26$$~~

$$SO = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 54 + 112 \\ 126 + 154 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{matrix} K \\ U \end{matrix}$$

SO $\rightarrow$ KU

$$LV = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \bmod 26$$

$$\begin{aligned}
 &= \begin{bmatrix} 3(11) + 8(21) \\ 7(11) + 11(21) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{matrix} T \\ W \end{matrix}
 \end{aligned}$$

LV $\rightarrow$ TW

$$\begin{aligned}
 ED &= \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 3(4) + 8(3) \\ 7(4) + 11(3) \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26} \\
 &= \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{matrix} K \\ J \end{matrix}
 \end{aligned}$$

ED $\rightarrow$ KJ

$\therefore$ Ciphertext : KUTWKJ